

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Use the classical electron energy $\hat{G}$ moment um relation for relativ istic de Broglie wavelength :

$\hat{\lambda} = h/(\hat{\lambda}(2mE)c)$,
where h is Planck $\hat{G}$ constant (in erg $\hat{A}$ ·s), m is the electron mass (9.11×10^{-28} g), c is the speed of light (3.00×10^{10} cm/s), and $E = 1 \text{ eV} = 1.60 \times 10^{-12}$ erg .

Calculate the factor :

$\hat{\lambda}(2mE/c^2) = \hat{\lambda}[(2 \times 9.11 \times 10^{-28} \times 1.60 \times 10^{-12})/(3.00 \times 10^{10})^2] \hat{\lambda} 1.875 \times 10^{-23} \text{ g}\hat{A}\cdot\text{cm/s}.$

Then $\hat{\lambda} = h/(\hat{\lambda}(2mE)c) = (6.625 \times 10^{-27} \text{ erg}\hat{A}\cdot\text{s})/(1.875 \times 10^{-23} \text{ g}\hat{A}\cdot\text{cm/s} \times 3.00 \times 10^{10} \text{ cm/s}) \hat{\lambda} 1.23 \times 10^{-7} \text{ cm}.$

Correct answer

